where $\mathbf{\Sigma} \in \mathbb{R}^{K \times K}$ represents the covariance matrix that captures the relationship between different sentences in the embedding space, $\mathbf{Z} = [\boldsymbol{z}_1, \boldsymbol{z}_2, \cdots, \boldsymbol{z}_K] \in \mathbb{R}^{d \times K}$ represents the embedding matrix of different sentences, $\mathbf{J}_d = \boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\top$ is the centering matrix and $\mathbf{1}_d \in \mathbb{R}^d$ is the all-one column vector. Then, the proposed EigenScore can be defined as the logarithm determinant (LogDet) of the covariance matrix,

$$E(\mathcal{Y}|\boldsymbol{x}, \boldsymbol{\theta}) = \frac{1}{K} \log \det(\mathbf{\Sigma} + \alpha \cdot \mathbf{I}_K) \tag{5}$$

Here, $\det(\mathbf{X})$ represents the determinant of matrix $\mathbf{X}$, and a small regularization term $\alpha \cdot \mathbf{I}_K$ is added to the covariance matrix to explicitly make it full rank. Since the matrix determinant can be obtained by solving the eigenvalues, the EigenScore can be computed as

$$E(\mathcal{Y}|\boldsymbol{x}, \boldsymbol{\theta}) = \frac{1}{K} \log(\prod_i \lambda_i) = \frac{1}{K} \sum_i^K \log(\lambda_i) \tag{6}$$

where $\lambda = \{\lambda_1, \lambda_2, \cdots, \lambda_K\}$ denotes the eigenvalues of the regularized covariance matrix $\mathbf{\Sigma} + \alpha \cdot \mathbf{I}$, which can be solved by Singular Value Decomposition (SVD). Eq. 6 shows that the hallucination degree of LLM's generation can be measured by the average logarithm of the eigenvalues. The conclusion is intuitive, as the eigenvalues of covariance matrix capture the divergence and correlation relationship between embeddings of different sentences. When the LLM is confident to the answers and K generations have similar semantic, the sentence embeddings will be highly correlated and most eigenvalues will be close to 0. On the contrary, when the LLM is indecisive and hallucinating contents, the model will generate multiple sentences with diverse semantics leading to more significant eigenvalues. The following remark is also provided to explain why the proposed EigenScore is a good measure of knowledge hallucination.

Remark 1. LogDet of covariance matrix represents the differential entropy in the sentence embedding space. Differential Entropy is the natural extension of discrete Shannon Entropy $H_e(X) = -\sum_X -p(x) \log p(x)$. The differential entropy $H_{de}(X)$ in continuous space can be defined by replacing the probability function with its density function $f(x)$ and integrating over x, i.e., $H_{de}(X) = -\int_x f(x) \log f(x) dx$. In principle (Zhouyin & Liu, 2021), for a multivariate Gaussian distribution $X \sim N(\boldsymbol{\mu}, \mathbf{\Sigma})$, the differential entropy can be represented as

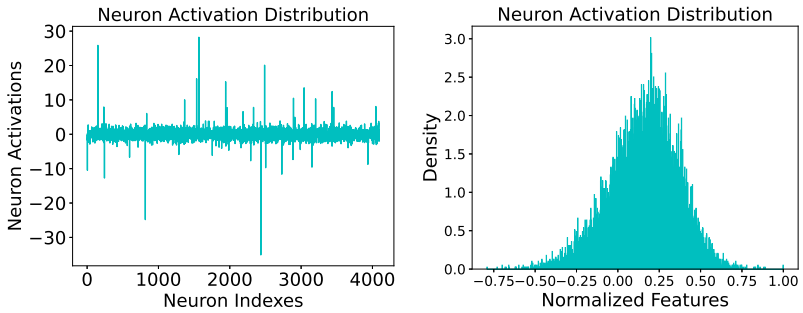
$$H_{de}(X) = \frac{1}{2} \log \det(\mathbf{\Sigma}) + \frac{d}{2}(\log 2\pi + 1) = \frac{1}{2} \sum_{i=1}^{d} \log \lambda_i + C \tag{7}$$

where d is the dimension of variables and C is a constant. Therefore, the differential entropy is determined by the eigenvalues (LogDet) of the covariance matrix.

According to **Remark 1**, the proposed EigenScore defined by Eq. 6 represents the differential entropy in the sentence embedding space, which offers valuable insight into using EigenScore as a semantic divergence measure. Compared to existing uncertainty or consistency metrics that obtained in logit or language space (Malinin & Gales, 2020; Huang et al., 2023; Lin et al., 2022), the advantages of EigenScore are: (1) It captures the semantic divergence (entropy) in the dense embedding space, which is expected to retain highly-concentrated semantic information compared to logits or languages (Reimers & Gurevych, 2019). (2) Representing semantic divergence in embedding space can effectively solve the semantic equivalence (linguistic invariances) problem (Kuhn et al., 2022) in natural language space. (3) Fine-grained semantic relationship among different responses can be exploited by using eigenvalues of covariance matrix. Therefore, through the exploration of dense semantic information in the internal states, the EigenScore is expected to outperform existing uncertainty and consistency metrics, resulting in improved hallucination detection performance.

3.2 Test Time Feature Clipping

Recent works have shown that the LLMs are subject to the risks of self-consistent (overconfident) hallucinations (Ren et al., 2022; Ji et al., 2023), which has not been considered by existing consistency based methods. Therefore, to address those failure cases caused by overconfident generation,



(a) Neuron Activation (b) Feature Distribution

Figure 2: Illustration of activation distributions in the penultimate layer of LLaMA-7B. (a) Activation distribution in the penultimate layer for a randomly sampled token. (b) Activation distribution for a randomly sampled neuron activation of numerous tokens.

a test time feature clipping approach is introduced during the computation of EigenScore. As shown in Figure. 2, we illustrate the activation distribution in the penultimate layer of LLaMA-7B. An intuitive observation is that the penultimate layer of LLMs tends to exhibit numerous extreme features, consequently increasing the likelihood of generating overconfident and self-consistent generations. Inspired by prior works that rectify internal activations to reduce overconfident prediction for Out-of-Distribution (OOD) detection (Sun et al., 2021; Djurisic et al., 2022; Chen et al., 2024), we introduce a test time feature clipping (**FC**) method to prevent LLMs generate overconfident hallucinations. To rectify those extreme features, the FC operation is defined as the following piecewise function

$$FC(h) = \begin{cases} h_{min}, & h < h_{min} \\ h, & h_{min} \leq h \leq h_{max} \\ h_{max} & h > h_{max} \end{cases} \tag{8}$$

where h represents the feature of the hidden embeddings in the penultimate layer of the LLMs, h_{min} and h_{max} are two thresholds for determining the minimum and maximum truncation activations. When $h_{min} = -\infty$ and $h_{max} = +\infty$, the output feature embedding is equivalent to the original output. For the determination of the optimal truncation thresholds, a memory bank which dynamically pushes and pops element in it, is utilized to conserve N token embeddings during test time. Then, for each hidden neuron, the thresholds h_{min} and h_{max} are set to the top and bottom p-th percentiles of the features in the memory bank. Refer to the three-sigma-rule Pukelsheim (1994), we set $p = 0.2$ in all cases. This implies that the activations falling within the largest and smallest top 0.2% in the memory bank are identified as abnormal features and subsequently truncated for reducing overconfident generation.

4 Experiments

4.1 Experimental Setup

Datasets. We utilize four widely used question answering (QA) datasets for evaluation, including two open-book conversational QA datasets CoQA (Reddy et al., 2019) and SQuAD (Rajpurkar et al., 2016), as well as two closed-book QA datasets TriviaQA (Joshi et al., 2017) and Natural Questions (NQ) (Kwiatkowski et al., 2019). We follow Lin et al. (2023) to utilize the development split of CoQA with 7983 QA pairs, the validation split of NQ with 3610 QA pairs and the validation split of the TriviaQA (*rc.nocontext subset*) with 9,960 deduplicated QA pairs. For the SQuAD dataset, we filter out the QA pairs with their flag *is_impossible* = *True*, and utilize the subset of the development-v2.0 split with 5928 QA pairs. The lengths of the sequences vary in the four datasets. Specifically, the ground truth answers in CoQA and SQuAD are relatively longer, while and TriviaQA typically consists of answers that are only with one or two words.

Models. We use two representative open source LLMs, including LLaMA (Touvron et al., 2023a) and OPT (Zhang et al., 2022) in our experiments. Specifically, we consider off-the-shelf LLaMA-7B [2], LLaMA-13B [3], OPT-6.7B [4] and their corresponding tokenizer provided by Hugging Face. We use the pre-trained wights and do not finetune these models in all cases.

Evaluation Metrics. Following prior work Kuhn et al. (2022); Ren et al. (2022), we evaluate the hallucination detection ability of different methods by employing them to determine whether the generation is correct or not. Therefore, the area under the receiver operator characteristic curve (AUROC) and Pearson Correlation Coefficient (PCC) are utilized as the performance measure. AUROC is a popular metric to evaluate the quality of a binary classifier and uncertainty measure (Ren et al., 2022; Lin et al., 2023). Higher AUROC scores are better. PCC is utilized to measure the correlation between the hallucination detection metric and the correctness measure, which is usually defined as the ROUGE score (Lin, 2004) or semantic similarity (Reimers & Gurevych, 2019) between the generated answers and ground truth answers. A higher PCC score is better.

Baselines. We compare our proposal with the most popular uncertainty-based methods **Perplexity** Ren et al. (2022) and Length-normalized Entropy (**LN-Entropy**) Malinin & Gales (2020), and the consistency-based metric **Lexical Similarity** (Lin et al., 2022). Besides, in order to investigate whether traditional OOD detection methods can be used for hallucination detection, we also introduce a popular OOD detection method **Energy** score (Liu et al., 2020) as a comparison method.

Correctness Measure. We follow Kuhn et al. (2022); Lin et al. (2023) to utilize both the ROUGE-L (Lin, 2004) and the semantic similarity (Reimers & Gurevych, 2019) as the correctness measure. ROUGE-L [5] is an n-gram based metric that computes the longest common subsequence between two pieces of text. The generation is regarded as correct when the ROUGE-L (f-measure) is large than a given threshold, which we set to 0.5 in our main experiments. Besides, we also use the embedding similarity as the correctness measure. The sentence embeddings of model generation and the ground truth answer are extracted by the *nli-roberta-large* model [6], and the generation is regarded as true when the cosine similarity between two embeddings is larger than 0.9.

Implementation Details. Implementation of this work is based on pytorch and transformers libraries. For the hyperparameters that are used for sampling strategies of LLMs' decoder, we set *temperature* to 0.5, *top-p* to 0.99 and *top-k* to 5 through the experiments. The number of generations is set to $K = 10$. For the sentence embedding used in our proposal, we use the last token embedding of the sentence in the middle layer, i.e., the layer index is set to int(L/2). For the regularization term of the covariance matrix, we set $\alpha = 0.001$. For the memory bank used to conserve token embeddings, we set $N = 3000$. When implement the Energy Score, we average the token-level energy score as the sentence-level energy score.

4.2 Main Results

Effectiveness of EigenScore. In Table. 1, we compare our proposed EigenScore with several representative reliability evaluation methods on three LLMs and four QA datasets. The results show that: (1) In both LLaMA and OPT models, our proposed EigenScore consistently outperforms other comparison methods by a large margin in CoQA, SQuAD and NQ datasets under different evaluation metrics. In particular, the EigenScore outperforms Lexical Similarity by 5.6% in CoQA and 8.9% in SQuAD with AUROC metric at most. (2) It's interesting to see that the Perplexity performs best in TriviaQA dataset but performs poorly on other datasets, especially for CoQA and SQuAD. This is because the generations and ground truth answers on TrivaiQA dataset is very simple, with only one or two words in the most cases. Therefore, the performance of different methods in TriviaQA is close and by simply averaging the token-level confidence as uncertainty measure performs well. (3) On average, the performance in LLaMA-13B is better than that in LLaMA-7B and OPT-6.7B, while the performances in LLaMA-7B is slightly better than that in OPT-6.7B. It demonstrates that better hallucination detection performance can be achieved with a more powerful pre-trained LLM.

[2]https://huggingface.co/decapoda-research/llama-7b-hf

[3]https://huggingface.co/decapoda-research/llama-13b-hf

[4]https://huggingface.co/facebook/opt-6.7b

[5]https://github.com/google-research/google-research/tree/master/rouge

[6]https://huggingface.co/sentence-transformers/nli-roberta-large